Trøndelagkonkurranse 2026 - Level I

Third Round - November 3, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Find all triplets (a, b, c) of natural numbers such that $a < b < c$ and

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \in \mathbb{N}.$$

Problem 2

A natural number $n > 1$ is called *agreeable* if an equilateral triangle T that can be divided into n smaller equilateral triangles such that:

- The sides of the smaller triangles are parallel to the sides of T.
- The smaller triangles do not overlap inside T. Vertices and edges can touch each other.
- Every portion of T is covered by a smaller triangle.

Show that 5 is not agreeable.

Problem 1

Since $a, b, c \in \mathbb{N}$ and $a < b < c$, we have that

$$1 \leq a, \quad 2 \leq b, \quad 3 \leq c,$$

so

$$1 \geq \frac{1}{a}, \quad \frac{1}{2} \geq \frac{1}{b}, \quad \frac{1}{3} \geq \frac{1}{c}.$$

Adding up, we obtain

$$1 + \frac{1}{2} + \frac{1}{3} \geq \frac{1}{a} + \frac{1}{b} + \frac{1}{c},$$

and since the left side is smaller than 2, the sum of the inverses of a, b and c can only reach 1.

Suppose $a \geq 3$, then $b \geq 4$ and $c \geq 5$, but this leads to

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq \frac{1}{3} + \frac{1}{4} + \frac{1}{5} < \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1,$$

which is not possible. Thus $a = 1$ or $a = 2$.

If $a = 1$, then

$$\frac{1}{1} + \frac{1}{b} + \frac{1}{c} = 1 \implies \frac{1}{b} + \frac{1}{c} = 0,$$

which is not possible since b and c are natural numbers.

If $a = 2$, then

$$\frac{1}{2} + \frac{1}{b} + \frac{1}{c} = 1 \implies \frac{1}{b} + \frac{1}{c} = \frac{1}{2}.$$

So far so good. If $b = 3$, then

$$\frac{1}{3} + \frac{1}{c} = \frac{1}{2} \implies \frac{1}{c} = \frac{1}{6} \implies c = 6$$

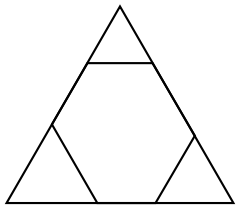
and we are done; $(a, b, c) = (2, 3, 6)$ is a triplet that satisfies the conditions of the problem. If $b \geq 4$, then $c \geq 5$ and

$$\frac{1}{b} + \frac{1}{c} \leq \frac{1}{4} + \frac{1}{5} < \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

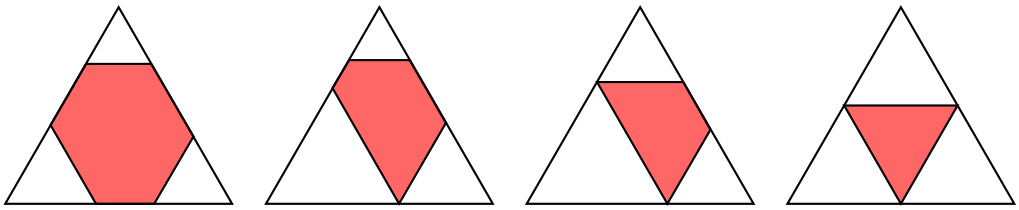
which is not possible. Hence, the only solution is $(a, b, c) = (2, 3, 6)$.

Problem 2

Suppose 5 is agreeable; then, since T has three vertices, we have to cover them with three different triangles, obtaining something like this:



Since the inner triangles can touch sides and vertices but not overlap, we have the following possible ways for these triangles



(a) (b) (c) (d)

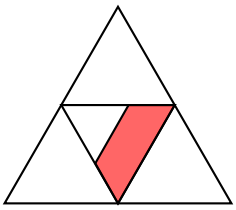
Distributed as the inner triangles have zero points of contact (a), one point of contact (b), two points of contact (c) and three points of contact (d).

In each case, T minus the triangles in the corner results in a hexagon, a pentagon, an isosceles trapezoid and an equilateral triangle, all of which are convex, and they need to be covered by two equilateral triangles.

Notice that the hexagon, the pentagon, and the isosceles trapezoid all have at least one angle of $120°$. This requires the two remaining equilateral triangles to touch at one vertex and share part of one side each. However, notice that if one triangle is bigger than the other one, we will have a concave pentagon, which is not allowed

Not allowed Allowed

Hence, the two remaining triangles have to be congruent and form a rhombus. This immediately rules out cases (a), (b) and (c) since this rhombus is neither a hexagon, a pentagon, nor an isosceles trapezoid. To rule out the last case (d), we can either cover the whole red area with one equilateral triangle, but this leaves one small triangle out, or partially cover one of the corners, resulting in something like



(d)

where the resulting red area will be an isosceles trapezoid, with four sides, so it cannot be properly covered by one equilateral triangle. Finally, we have ruled out all the possible cases, concluding that we **cannot** perfectly cover an equilateral triangle with 5 smaller equilateral triangles, so 5 is not agreeable.